

Why clause structure is judged like tense and agreement: public-update operators and grammaticality

Abstract

Grammaticality judgments are selective: speakers reject some mismatches as ungrammatical while treating accent, register, and many lexical choices as negotiable. This article argues that the contrast tracks public-update operators: restricted form–value relations that configure what move an utterance counts as, who is committed to what, and which responses are licensed. Clause typing, argument linking, polarity, evidentiality, and grammaticalized tense–aspect occupy this role where they belong to closed, accountable repertoires. The account explains why clausal architecture patterns with tense and agreement, why tone, gesture, and lexical items can become grammatical, and why ideological policing differs from first-order operator mismatch. It predicts repair, preemption, and processing effects around operator contrasts cross-linguistically.

Running head: Public-update operators and grammaticality

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I THE PUZZLE: WHY “GRAMMAR” PICKS OUT SOME THINGS AND NOT OTHERS

The asterisk is selective. Some mismatches are treated as straightforward errors of “grammar”, often with an accompanying sense that there’s nothing to debate: wrong agreement, illicit clausal morpho-syntax, incompatible tense–aspect morphology, missing obligatory evidentials, or mis-selected question particles in systems where such particles are required. Other mismatches are treated as infelicities of social positioning: an accent that indexes the wrong stance, a style that fails to fit the activity type, a form that sounds “too formal” or “too intimate”. Still other mismatches are treated as mere lexical choice: suboptimal perhaps, but not illicit.¹

A familiar temptation is to reduce this to a contrast between “structure” and “use”: morpho-syntax counts as grammar; accent and lexical choice don’t.² But this gloss is neither typologically adequate nor explanatory. Many languages use prosody or tone as obligatory exponents of grammatical contrasts (Hyman, 2011; Ladd, 2008). Many grammatical categories aren’t universal: a language may lack tense (Comrie, 1985) or number (Corbett, 2000) while sustaining robust categorical judgments elsewhere. And some lexical items enter operator contrasts, with constraints indistinguishable from inflectional morphology (Bybee, 2010; Hopper & Traugott, 2003).

¹ Speakers do label accent and dialect variation as “bad grammar” for social reasons, but the interactional repair profile differs, a point taken up in §8.

² Introductory diagrams draw a reassuring boundary; languages then spend the rest of the semester crossing it.

The core question is not why morphosyntax is insulated from meaning, but why some form–value relations become non-negotiable infrastructure, policed as membership facts, while others remain negotiable resources for stance, identity, and content selection.

The argument developed here is that this difference tracks a difference in *VALUE*: what a contrast does in a system, what it rules in or out for uptake.³ Consider polarity: choosing *not* over affirmative changes what inferences are licensed and what responses are relevant. Or clause type: a question particle vs. a declarative ending changes what kind of move the utterance counts as. These are *OPERATOR* contrasts, and grammaticality judgments, I’ll argue, target operator value.

2 WHY SOME CONTRASTS ARE PUBLICLY ACCOUNTABLE

The starting point is a deliberately non-idealized view of grammar. A community’s linguistic system isn’t merely a set of strings or derivations, but a repertoire of conventionalized form–value relations that enable coordination in interaction.

Consider two misfires. In the first, a speaker produces a clause with the wrong tense or a mismatched agreement marker; the hearer responds “Wait, who did what?” or “Did you mean X or Y?” Repair targets the structure: the update couldn’t go through. In the second, a speaker uses an accent or register that doesn’t fit the setting; the hearer understands the utterance perfectly but reacts to the speaker’s social positioning: “Why are you talking like that?” Uptake succeeds; what’s targeted is stance or persona, not the move itself.

This contrast reflects what I’ll call *PUBLIC ACCOUNTABILITY*. The relevant sense of public update is the one captured by work on common ground and grounding: interlocutors don’t merely produce signals; they attempt to make commitments mutually recognizable and mutually ratifiable (Clark, 1996; Clark & Brennan, 1991; Stalnaker, 1978, 2002). Some aspects of linguistic form are publicly accountable in this sense; others aren’t.

Repair practices make this selectivity visible. When an utterance isn’t understood, or is understood under a suspect analysis, interlocutors systematically mobilize repair mechanisms (Schegloff et al., 1977). These mechanisms presuppose that some dimensions of form are designed to be recognizably “the same thing” across tokens and across speakers, because they’re the dimensions on which the interactional work turns.

Public accountability isn’t the same as public consequence. Indexical meaning is real and systematic, but it’s typically tolerated as variable, defeasible, and contextually negotiable. Accent and style are central to social meaning, but they aren’t normally treated as licensing conditions on what an utterance can “count as” in the way that clause-typing, polarity, or evidential marking can be (Agha, 2007; Coupland, 2007; Eckert, 2008, 2012). Lexical choice is similarly accountable in an ethical or epistemic sense (one can be challenged for using a slur, or for choosing a misleading term), but that accountability differs from “this form isn’t in the repertoire”.

If *VALUE* is always relational and contrastive, the relevant question is which contrasts a community treats as foundational for public update in interaction. Those contrasts constitute what I’ll call the *OPERATOR STRATUM*.

³Here *VALUE* has the Saussurean sense of *valeur* (de Saussure, 1916): the identity of a linguistic unit is constituted by its position in a system of contrasts (what it patterns with, what it opposes, what interpretations it makes available). It isn’t to be confused with feature-value assignment or decision-theoretic value.

3 THE OPERATOR STRATUM

3.1 THE OPERATOR-STRATUM HYPOTHESIS

Public-update operators are restricted form–value relations that signal what kind of move a speaker is making and how that move is to be taken up. I use the label narrowly: not as a taxonomy of everything called an operator, and not as a claim that every formal operator in a grammar has the same interactional role. The proposal is a hypothesis about which form–value relations fall under the accountability regime targeted by grammaticality judgments. The starting point is not the judgment itself, but two independently observable dimensions: what kind of update the relation licenses, and how the relation is organized in the community’s repertoire.

(I) PUBLIC UPDATE CONDITION. The relation conventionally contributes to how an utterance updates shared commitments or allocates interactional roles (call it common-ground bookkeeping), but in a specific way: it supplies an *instruction* for the update, constraining what counts as an appropriate next move (Farkas & Bruce, 2010; Murray, 2014; Stalnaker, 2002). Clause-type, illocutionary force, and evidential anchoring are paradigm cases: they tell hearers how to file the contribution. In Murray (2014)’s terms, many operator contrasts are structuring updates rather than ordinary common-ground additions: they configure the discourse context in which at-issue content can be entered, challenged, or elaborated. A vivid adjective or slang term, by contrast, contributes content but doesn’t supply an update instruction in this sense.

The condition identifies licensed update types rather than successful updates. Truth knowledge evaluates whether a licensed update succeeds; contextual anchoring can determine which operator value is licensed. A speaker can say *It’s true* of a false situation: the assertion fails epistemically, but affirmative polarity and declarative force are licensed. By contrast, *Yesterday the bill passed* selects a past-time anchor; **Yesterday the bill pass* missets the tense value in varieties where finite past morphology is obligatory.

Contextual anchoring is constrained. Context selects among independently licensed operator values, constructional frames, and ellipsis patterns; it doesn’t create a form–value relation ad hoc. *I have five years* can succeed as an elliptical reply to “How long do you have left?” because the frame supplies the relevant duration. The same reply to “How old are you?” fails in ordinary English because age is expressed through the *be* frame, not because the hearer lacks the intended meaning.

(II) REPERTOIRE CONDITION. The relation belongs to a restricted, conventionalized contrast set whose values are recurrently selected in describable structural environments, whose alternatives are mutually exclusive in those environments, and whose extension is not freely innovated token by token. Tense paradigms, agreement systems, and complementiser inventories are paradigm cases. Idioms and fixed collocations are near misses: they may be rigid, but they don’t serve as combinatorial values in a contrastive system.⁴

Repeated use matters because it turns a licensing pattern into an expectation. Speakers encounter comparable contexts in which a restricted choice set is available: one operator value is selected, incompatible values are excluded, and the pattern projects forward to future tokens. That recurrence explains how speakers can detect operator mismatches without making detector output one of the membership conditions. Stability alone remains insufficient: phonotactics, collocations, and enre-

⁴These conditions and the coordination pressures that sustain them are mutually constitutive; actual form–value relations approximate them to varying degrees.

gistered styles can be stable over time while falling outside the operator stratum unless they also satisfy the public update condition.

Operator contrasts can be morphological, syntactic, phonological, prosodic, gestural, or lexical in form. What matters is their conventional role: they supply closed-system values for interpretation and interaction. Morphology and clausal architecture are privileged *where they instantiate operator contrasts*, not because they form a separate module.

The term OPERATOR is continuous with, but not identical to, its use in Role and Reference Grammar, Functional Grammar, and Functional Discourse Grammar. That tradition treats categories such as aspect, tense, modality, negation, and illocutionary force as operators over layers of clause structure, and it provides a structural-functional inventory of operator categories and scopal domains (Bentley et al., 2023; Hengeveld, 1989; Hengeveld & Mackenzie, 2008; Van Valin, 2005; Van Valin & LaPolla, 1997). The operator-stratum account shifts the explanandum. It asks why mismatches in some such form–value relations are experienced and policed as grammaticality mismatches. Operator-stratum membership here is therefore not only a scopal or morphosyntactic relation, but a role in public accountability: the operator value supplies a conventional instruction for update, uptake, commitment allocation, or role tracking.

Operators are intensely contrastive, but at a different functional level than phonemes. Phonemic value is primarily discriminative: its semiotic job is to keep signs apart. Operator value is contrastive at the level of interactional control: it selects a publicly accountable instruction for public update (clause type, polarity, evidential anchoring, and so on). Both are paradigmatic, but they make different contributions to the system.

Analogically, operators are header-like rather than payload-like. They specify how an utterance is to be taken up (what move it counts as, who’s committed to what, how subsequent contributions are constrained) rather than contributing the conceptual substance of what’s being communicated. This yields three strata of linguistic infrastructure: expression-shape constraints that make an utterance recognizable as a token of the system, operator contrasts that configure public update, and payload resources that contribute negotiable content or stance. Their mismatch profiles differ accordingly. Expression-shape mismatches block recognizability; operator mismatches block, distort, or reclassify the intended public update; payload mismatches invite negotiation.

Stratum	Contribution	Mismatch profile	Example
Expression-shape	Recognizable token	“Not a word”	phonotactic mismatch
Operator	Update instruction	“Can’t say that as that move”	obligatory operator mismatch
Payload	Content/stance	Negotiation, not rejection	lexical mismatch

Table 1: Three strata of linguistic infrastructure and their characteristic mismatch profiles.

The central puzzle concerns the middle stratum: why clausal architecture clusters with tense, number, and agreement as a target of grammaticality talk, rather than with accent or lexical choice. The answer developed below is that clausal architecture behaves “like grammar” when it packages operator values for update, role allocation, and uptake.

3.2 BOUNDARY CASES

Public consequence isn’t enough for operator-stratum membership. Many forms affect participant commitments, how they’re evaluated, or what repairs are available. The operator claim is narrower: a

form–value relation is an operator contrast only when it both configures a public-update instruction and belongs to a restricted, accountable repertoire. Table 2 makes the boundary explicit.

Case	Public?	Restricted?	Update instruction?	Predicted mismatch profile
Tense/evidential marker	yes	yes	yes	repair targeting operator value
Question particle	yes	yes	yes	uptake-role confusion
Slur/taboo item	yes	partly	no, except as social act	sanction of act or speaker
Proper-name misselection	yes	open-ish	no	referential repair
Phonotactic mismatch	yes	yes	no	recognizability problem
Register mismatch	yes	gradient	usually no	stance or social evaluation
Interjection token	yes	partly	sometimes	uptake configuration without operator-stratum membership
French determiner allomorphy	yes	yes	no/edge	form-selection mismatch at a stratum boundary

Table 2: Boundary cases for public-update operators. Public consequence alone doesn't define operator-stratum membership.

Three boundary cases show why both conditions are needed.

Slurs satisfy the public-consequence test but usually fail the operator test. A racial slur used as a vocative, or a misogynistic slur used as a predicate, can radically alter the social action performed and can make the speaker answerable for that action. But the utterance normally remains structurally fileable as an address, assertion, accusation, or description. The problem is that the move is morally or socially impermissible, not that the hearer lacks an instruction for how to update the discourse. The predicted response is therefore sanction of the act or speaker, not repair of an operator value.

Interjections sit closer to the operator stratum. They can configure uptake, stance, attention, repair, or floor management (Ameka, 1992). Related work on English interjections observes that *oh* can mark a change in epistemic state, *huh?* can initiate repair, *mmhm* can acknowledge or continue a turn, *uh* can delay while holding the floor, and *wow* can display surprise or assessment (Author, 2026d).

But many interjection tokens are not selected from a closed set of mutually exclusive values in the way tense, polarity, or clause type are. They can be repeated, stretched, blended, innovated, or substituted without producing a categorical licensing failure. In those cases, an odd interjection may disturb uptake or stance without leaving the utterance outside the operator repertoire. If an interjection-like form is conventionalized into a closed, distributionally accountable paradigm, it can enter the operator stratum; the point is that uptake configuration alone is not enough.

French **le hiver* tests the other side of the boundary. The mismatch isn't generic phonology or accent. It fails morphophonological allomorph selection inside a closed determiner/clitic repertoire: before vowel-initial *hiver*, the expected alternant is *l'hiver*. That repertoire also participates in reference setting, so the case lies at the expression-shape/operator boundary. The asterisk is warranted because a closed form-selection pattern has been violated; the lesson is that a superficially phonological mismatch can attract grammaticality talk when it's embedded in an accountable repertoire.

3.3 DIAGNOSTICS

The operator stratum can be diagnosed empirically across languages.

PARADIGMATIC CLOSURE. Operator contrasts typically occupy relatively small, enumerable paradigms: polarity contrasts, evidential sets, tense–aspect systems, agreement paradigms, switch-

reference markers, question particles, complementiser inventories, and so on. Open-class lexical sets don't behave this way.

BROAD SCOPE. Operator contrasts typically configure the update potential of an utterance as a whole, affecting argument linking, scope, clause type, and what responses are licensed. In public-update terms: they constrain which next moves are relevant (clause type), who's committed to what (evidential anchoring, argument structure), and how the contribution can be challenged or elaborated (dependency structure, polarity).

OPPORTUNITY MASS AND PREEMPTION. Many operator contexts are encountered constantly in ordinary interaction, which produces strong preemption dynamics. Because *went* is everywhere, **goed* gets treated as flatly non-licensing; because *depend on* wins every time, **depend of* can't gain a foothold. Left-branch extraction makes the same point for clausal architecture: English speakers repeatedly need to ask about a modifier–noun package, but *Which car did you buy?* occupies that operator context while **Which did you buy car?* remains systematically untaken (Author, 2026a). The gap is syntactic, high-opportunity, and semantically transparent: the problem isn't morphology or lexical selection, but non-licensing in an operator context. A competitor that systematically wins in a large opportunity set quickly drives a rival option toward non-licensing (Bybee, 2006, 2010).⁵ Where opportunity mass is lower, the same kinds of contrasts may still belong to the operator stratum, but categoricity and preemption effects are predicted to be weaker and more register-bound.

NEUROCOGNITIVE SEPARATION IN RESPONSE PROFILE, NOT IN MODULE. The proposal is compatible with domain-general accounts of prediction and error signalling, but it predicts that operator mismatches should systematically elicit distinct response profiles from conceptual anomalies, because operator mismatches interfere with the mapping from form to public update, rather than merely producing a surprising concept combination. This aligns with well-known ERP dissociations between semantic incongruity and morphosyntactic anomaly (Friederici, 2011; Kutas & Hillyard, 1980; Osterhout & Holcomb, 1992) without implying that syntax is autonomous.

Closure, scope, and opportunity structure identify operator contrasts independently. Categorical judgments, repair profiles, and neurocognitive response profiles are predicted ways those contrasts should behave, not membership criteria. Because the diagnostics track contribution rather than substance, a grammaticalized contrast can become relevant to grammaticality talk even when the exponent is tonal, prosodic, or gestural.

4 WHY CLAUSAL ARCHITECTURE BELONGS TO THE OPERATOR STRATUM

Clausal architecture is often discussed as if it were a matter of arranging words. That's an English-biased picture. Cross-linguistically, clausal architecture is realized through diverse operator exponents: affix order in polysynthetic systems, templatic morphology, clitic clusters, tone melodies, non-manual marking, rigid word order, flexible order coupled with case, and various mixtures. The relevant generalization is not linearization but operator packaging.

The thesis is narrower than “all syntax belongs to the operator stratum”. Clausal architecture belongs with tense and agreement when it realizes publicly accountable operator contrasts: clause type, argument linking, dependency scope, and reference tracking. Some syntactic constraints may instead reflect expression-shape regularities, processing limits, or constructional conventionalization.

⁵The effect is Bayesian: the strength of the inference against the absent form tracks the frequency with which the licensed competitor occupies the relevant operator context.

The syntax most readily experienced as grammatical mismatch is predicted to be operator-rich, but not every syntactic asterisk is an operator mismatch.

The relevant stability is projectible rather than merely frequent (Author, 2026b). In clause typing, argument linking, dependency scope, and reference tracking, speakers encounter recurrent environments in which a small set of values is available, one value is selected, and incompatible alternatives are excluded. That projectible licensing profile is what makes a local clausal mismatch feel like a failure in the shared repertoire rather than an isolated processing difficulty or pragmatic misfire.

Clausal architecture packages operator contrasts in at least four domains.

CLAUSE TYPE AND RESPONSE SPACE. Interrogatives, imperatives, declaratives, and related types encode different update potentials and different norms for what counts as a relevant next move. The contrast is interactionally basic, and languages grammaticalize interrogatives using heterogeneous resources (particles, morphology, intonation, or combinations) (Dryer, 2013; Ladd, 2008). Where a language requires an interrogative exponent for the intended question, declarative morphology leaves the addressee to decide whether to answer or merely acknowledge; the intended public update is unsettled.

ARGUMENT LINKING AND PARTICIPANT ROLES. Clausal organization encodes which participant is presented as actor, undergoer, experiencer, and so on. The particular implementation varies typologically, but the operator contrast is projectible: it constrains the space of recoverable role assignments, and it does so in a publicly accountable way. If the wrong role assignment is signalled, the hearer tracks the wrong participant as agent or patient, derailing downstream inference.

DEPENDENCY MANAGEMENT AND SCOPE. Relative clauses, complement structures, and long-distance dependencies aren't simply compositional ornaments. They package how information is nested, attributed, and scoped. If scope relations are unclear, the hearer doesn't know what the speaker is committed to vs. merely reporting or considering; these are conditions on what an utterance publicly commits to and how it can be challenged (Farkas & Bruce, 2010; Stalnaker, 2002). Ordinary English complement choice makes the point: *Kim knows that Pat left* presents the embedded proposition as available for commitment, *Kim wonders whether Pat left* opens a question, and **Kim wonders that Pat left* misselects the complement type for the intended update. Put differently, grammaticality asks more than whether dependency arrows can be drawn; it asks whether the dependency configuration implements a licensed update in the relevant operator repertoire (Gibson, 2026).

REFERENCE TRACKING ACROSS CLAUSES. Many languages treat cross-clausal coherence as an operator contrast. Switch-reference is a textbook case: a morphosyntactic system signals whether the subject of one clause is the same as or different from that of another (van Gijn, 2016). A wrong marker licenses the wrong tracking relation or leaves tracking unresolved, which can derail the whole multi-clause sequence. The key point here isn't the particular pivot (which varies) but the fact that a community can grammaticalize the tracking problem as operator value.

To illustrate the form of the claim, consider Amele, a Papuan language in which dependent switch-reference verbs in clause chains mark whether the relevant argument is maintained or changed across adjacent clauses (Roberts, 1987, 1988, 2023). Roberts's examples contrast forms such as *bu-m-ig* "come-SS-1SG" and *ho-co-min* "come-DS-1SG" before a final tense-marked verb. The operator contrast isn't "a content word"; it supplies a clause-linking operator value. Operator mismatches are predicted to be treated as structural mismatches: the wrong marker licenses the wrong clause-linking update, or leaves the intended update unresolved.

A second, different kind of illustration comes from Japanese sentence-final particles. In many registers, *ka* conventionally marks interrogative force, while declaratives are unmarked or carry other particles; intonation interacts with these choices, and colloquial speech allows more variation (Kuno, 1973; Shibatani, 1990). The particle doesn't add much conceptual content. Rather, it helps determine what kind of move the utterance counts as, how it's to be taken up, and what responses are licensed. The relevant point is membership in a clause-typing repertoire, not automatic failure. An omitted *ka* isn't ungrammatical just because an interrogative interpretation is intended: it may yield a declarative construal, a colloquial intonation-marked question, or some other licensed move. *Ka* becomes relevant to ungrammaticality only where the repertoire makes a particular value obligatory, excludes a competing value, or blocks a combination with other particles, intonation, or morphosyntax. In that sense, it illustrates how a small, conventionalized contrast can belong to grammar because it configures public update, while still leaving room for register-sensitive and construction-specific variation.

Together, the Amele and Japanese cases separate two claims that are easily conflated. Some clausal operators generate clear structural mismatches when the wrong value is selected; others, like *ka*, show operator membership without automatic failure. In both cases, clausal architecture matters because it packages values for public update, not because syntax is insulated from interaction.

A familiar morphosyntactic reflex of the operator/payload partition is the complement–modifier contrast. Many head–dependent relations are publicly accountable because they fix argument linking, attribution, or scopal organization; these are complement-like in the relevant sense even when their exponents are PP- or clause-sized. English *put* shows the operator-like side: *I put the book on the table* supplies a place value the construction requires, while **I put the book* leaves the clausal instruction incomplete. By contrast, *I read the book on the train* uses a PP to enrich payload; omitting it changes content but doesn't break the clausal instruction.

This recasts some so-called obligatory adjuncts as cases where a construction conventionalizes an operator contrast whose value needs to be supplied overtly or implicitly. It also predicts that some modifier-shaped material (negation, focus particles, degree operators) will pattern with the operator stratum where their licensing conditions make one value obligatory or exclude another, despite not looking “argumental”.⁶

5 WHY TENSE AND NUMBER BEHAVE SIMILARLY WITHOUT BEING UNIVERSAL

On this account, tense and number behave “like grammar” in languages where they're operator contrasts, and are irrelevant to grammaticality talk where they're not grammaticalized. This avoids a common slippage between “salience in familiar European languages” and “cross-linguistic necessity” (Haspelmath, 2007).

Tense marking is an especially transparent example. Where tense is grammaticalized, it's an operator contrast for temporal anchoring and public update, and operator mismatches are often treated as licensing failures (Comrie, 1985). Where tense isn't grammaticalized, temporal anchoring is managed through other operator contrasts (aspect, evidential access, discourse particles) and through contextual inference. The proposal predicts that speakers in the latter case shouldn't experience missing tense marking as “ungrammatical”, because there's no operator contrast to mismatch.

⁶Nothing here hinges on a particular complement/adjunct diagnostic; the point is that dependencies that supply operator values and payload-enriching dependencies tend to diverge in repair profile and licensing expectations.

Number behaves analogously. In languages where number is obligatory in agreement or nominal morphology, it's part of the operator repertoire for tracking reference in public commitments, and a number mismatch is a paradigmatic licensing failure. In languages where number is optional or limited, other contrasts fill the reference-tracking role, and number mismatches have different status (Corbett, 2000).

Operator repertoires vary across languages; grammaticality judgments target operator contrasts *wherever they occur*.

6 WHY ACCENT AND LEXICAL CHOICE ARE USUALLY DIFFERENT

Accent and lexical choice typically have different kinds of value.

6.1 INDEXICAL VALUE AND NEGOTIABILITY

Accent, style, and register contribute indexical value: they position speakers relative to social categories, stances, and activity types (Agha, 2007; Coupland, 2007; Eckert, 2008; Silverstein, 1976). Indexical value is often rich and structured, but it isn't typically implemented as a small closed paradigm with obligatory selection conditions tied to public update. It's negotiable in ways operators aren't. Interlocutors can reinterpret an accent shift as play, accommodation, quotation, or stance work without treating it as a licensing failure. The accountability regime is different: style can be challenged, but it isn't ordinarily policed as "not in the repertoire".

Indexicality can be obligatory in social practice without becoming an operator licensing condition. Much indexical meaning is *meta-pragmatic*: it becomes salient through social ideologies and enregisterment processes (Agha, 2007; Eckert, 2012). That makes it both powerful and variable, but it rarely supplies an operator value that has to be selected for the utterance to count as a recognized move.

6.2 CONCEPTUAL VALUE AND OPEN CLASSES

Open-class lexical items primarily contribute conceptual content, and the system is designed to be expandable and to tolerate innovation. Lexical choice can be infelicitous, misleading, or socially harmful, but it's seldom treated as a failure of combinatorial licensing. Lexical innovation is one of the routine mechanisms by which communities extend their repertoire, and it typically proceeds without the categorical judgment profile associated with operator mismatches.

"Open-endedness" has two dimensions that matter here. Lexical inventories are open in the sense of expandable type-sets: innovation is normal and the system tolerates new entries. Compositional semantics is open-ended in a different sense: unbounded productivity from combining a finite stock. Operators lack the first kind of openness (paradigmatic closure is one of the diagnostics), but they participate fully in the second: operator values compose with payload. Negation scopes, evidentials anchor commitments, and clause type constrains response space. Because the inventory is closed and the contribution is control-like, the community can stabilize sharp licensing expectations around them.

That difference explains why "wrong word" and "wrong clausal form" often feel qualitatively different even when both are understood. A wrong operator value threatens the public recognizability of the intended update; a non-optimal lexical choice typically doesn't.

6.3 CROSS-SUBSTANCE OPERATORS

Operator status is a role, not a substance. Phonological material can be an operator exponent, and lexical items can enter operator contrasts when they grammaticalize.

Systems in which tone marks tense, focus, or polarity are paradigmatic cases of phonological material realizing operator value, and they should attract operator-directed repair where the relevant value is obligatory or mutually exclusive (Hyman, 2011). Intonational contours that conventionalize clause type are operator exponents; when they're required or exclusionary, they should pattern like question particles rather than like accent (Ladd, 2008). This aligns with work on rising declaratives and intonational commitments, where intonational tunes modify context-update functions rather than merely colouring pronunciation (Gunlogson, 2003; Rudin, 2022).

Signed languages make the same point in another modality. In American Sign Language, a yes/no question can be marked by nonmanual cues such as brow raise over the relevant clause. The operator value is carried by scopal nonmanual marking, not by a segmental word, but it still configures clause type and response space (Sandler & Lillo-Martin, 2006).

Lexical items can enter operator contrasts through grammaticalization. Paradigms shrink, content fades, and licensing constraints emerge (Bybee, 2010; Hopper & Traugott, 2003). Many "lexical" items already participate in operator contrasts without diachronic grammaticalization: function words, particles, auxiliaries, complementizers, and polarity items occupy restricted paradigms, show distributional licensing, take broad scope, and attract repair sensitivity. English preposition selection illustrates the split. Some prepositions participate in strictly licensed head-dependent relations that set argument linking; others remain payload choices whose selection depends on spatial or conceptual content. They're lexical in form but operator-like when they occupy the first role.

Phonotactics sharpens this point. Phonotactic constraints are repertoire-like: they're conventionalized, relatively stable, and mismatches are often treated categorically ("not a possible word"). But they don't satisfy the public-update condition. What they regulate is whether a token can be recognized as a well-formed expression-shape, not whether it specifies an update instruction correctly. The metalinguistic reaction profile is correspondingly different: operator mismatches invite "you can't say that (as a move)"; expression-shape mismatches invite "that's not a word / what did you say?"; lexical and pragmatic mismatches preserve recognizability but invite negotiation about content, stance, or appropriateness.

6.4 AN INFORMATION-THEORETIC FRAME

Operators don't usually add much payload content in themselves; they supply operator values that narrow how the rest is interpreted. Once an utterance is marked interrogative, the relevant next moves narrow to a small menu: answer, decline, request clarification. Declaratives license a different menu. Polarity flips entailment relations, changing what follows as a reasonable inference; truthfulness bears on update success rather than polarity licensing. Case and agreement constrain role assignment. In each case, a small, closed-paradigm operator value causes large downstream consequences for interpretation and uptake.

An information-theoretic perspective sharpens this point.⁷ Operator values are drawn from small

⁷Different linguistic subsystems trade off contrast-set size, within-type variability, and contextual predictability differently. Phonemes have a small contrast set but high within-category variability (the classic "lack of invariance" problem (Lieberman et al., 1967)). Lexemes occupy a huge contrast set but manage it by distributing information across sequences

contrast sets with constrained exponents. The hearer's relevant uncertainty isn't primarily over the marker itself, but over public-update states: assertion vs. question vs. command, self-commitment vs. attributed commitment, one role assignment vs. another, one challenge condition vs. another. An operator value conditions the downstream distribution over licit next moves, role assignments, and commitment states. The entropy reduction is therefore downstream; it need not equal the marker's local surprisal. A wrong value can have disproportionate coordination costs even if it's low-information in the Shannon sense (Cover & Thomas, 2006; Shannon, 1948).

This explains a puzzle: grammatical marking is often locally redundant.⁸ Mahowald et al. (2023) show that across 30 languages, subjecthood in simple transitive clauses can usually be inferred from lexical content alone. Overt grammatical cues are redundant in a majority of everyday cases, but crucial for the minority where lexical semantics doesn't settle the roles (reversible events, non-prototypical assignments). Operators can be locally redundant but globally update-relevant. Their contribution isn't to maximize information in each occurrence, but to stabilize a channel whose values remain available when context, lexical semantics, or discourse expectations underdetermine the update. Operators are maintained even when locally predictable because redundancy buys robustness and supports the communication of rare or reversible meanings.

The same logic explains why grammatical material can be phonetically reduced. High-frequency, high-predictability items (often function words⁹ and inflections) undergo reduction precisely because context carries the load (Jaeger, 2010). Operators can tolerate variable realization without losing operator-stratum membership, because what matters is their role in a small, closed paradigm, not the phonetic stability of their exponents.

This information-theoretic picture connects directly to opportunity mass. Dense exposure stabilizes preemption dynamics and sharpens licensing boundaries (Bybee, 2006, 2010). Once a community converges on a small, high-frequency paradigm with large downstream entropy effects, deviations become easier to detect and sanction.

7 TYPOLOGICAL CONSEQUENCES: WHAT VARIES IS WHICH CONTRASTS ENTER THE OPERATOR STRATUM

The operator stratum is a comparative concept: it identifies an interactional role that can be realized by different structures in different languages (Haspelmath, 2007). Typological diversity then concerns which contrasts enter that role, not whether English categories are universal.

Four domains are particularly instructive.

7.1 INSTRUCTIVE DOMAINS OF VARIATION

EVIDENTIALITY AND EPISTEMIC AUTHORITY. If evidentiality is obligatory and closed-paradigm in a language, then evidential mismatches should pattern like tense or agreement mismatches: categorical judgments, repair targeting the operator value (Aikhenvald, 2004). Where evidentiality isn't grammaticalized, similar meanings are available through lexical or periphrastic means, but they can be challenged as deceptive or inappropriate without being treated as structurally illicit. The diagnostic

and context; word length tracks average information content better than raw frequency (Piantadosi et al., 2011).

⁸Engineers call this redundancy; linguists call it agreement.

⁹Here, FUNCTION is reserved for its sense in *The Cambridge Grammar of the English Language*: the syntactic role an element plays (determiner, modifier, complement, etc.) or the established term FUNCTION WORDS. What operators *do* (their semiotic work) is referred to as their CONTRIBUTION.

is whether the mismatch changes the licensed update or merely raises questions about the speaker's reliability.

EGOPHORICITY AND PERSPECTIVE MANAGEMENT. If a language grammaticalizes perspective or ego-involvement as a closed paradigm, then perspective mismatches should pattern like agreement mismatches where the paradigm makes one value obligatory or excludes another: repair targets the operator value. Tibetic languages and others have been argued to do exactly this (Floyd et al., 2018; Tournaud, 2008). Regardless of the preferred analysis, these contrasts are prime candidates for operator-stratum membership: they govern whose epistemic position is encoded as the relevant anchor for a clause's update potential.

Where languages do *not* recruit perspective into the operator stratum, similar meanings remain expressible but are policed differently, as pragmatic misalignment rather than licensing failure.

PHONOLOGICAL EXPONENTS VS. ACCENT. If tone marks tense, aspect, or polarity in a language, then tonal mismatches should pattern like morphosyntactic mismatches when the intended operator value is fixed by the construction or context, not like accent shifts. That prediction makes tonal languages a crucial test case (Hyman, 2011). A language may distinguish tense values primarily through tonal melodies on the verb (one pattern for remote past, another for recent past), with minimal segmental differentiation between the forms.

A concrete case is Ngwi, a West-Coastal Bantu language whose Tense–Aspect–Mood–Polarity system includes grammatical tone in a compact set of verbal conjugations (Pacchiarotti & Bostoen, 2024). In such systems, tone isn't merely phonetic dress on a segmental form; it can be part of how an operator value is realized. A speaker who produces the wrong tone in these contexts need not be heard as merely “accented”; they may instead be heard as having selected the wrong tense, aspect, mood, or polarity operator value, inviting repair of the “did you mean X or Y?” type. This contrasts with sociolinguistic variation in the same languages, where tonal realization may shift to index region or age without affecting the operator value. The diagnostic: does the mismatch change what move was made, or just who made it?

SWITCH-REFERENCE AS INSTRUCTION. Switch-reference systems exemplify how clausal architecture can package an operator contrast beyond simple word order. In Amele, for example, a dependent verb in a clause chain can mark same-subject vs. different-subject linkage before the final tense-marked verb (Roberts, 1987, 1988, 2023).

- (1) a. *Verb-SS.SEQ* ... *Verb* (instruction: keep same participant active)
- b. *Verb-DS.SEQ* ... *Verb* (instruction: switch participant role)

This marker choice is an explicit instruction for reference tracking: it constrains the assignment of participant roles in the upcoming increment. An operator mismatch (using the same-subject marker when the subjects are different) disrupts reference assignment and clause linkage; repair targets reference tracking, not stance or style.

7.2 THE BOUNDARY CASE: T/V AND GRAMMATICALIZED SOCIAL DEIXIS

If an indexical distinction like addressee status becomes obligatory and paradigmatically closed, then some mismatches should pattern less like free register choice and more like operator mismatches. T/V distinctions (honorific pronouns and agreement) test this prediction (Brown & Gilman, 1960; Helmbrecht, 2013). French *tu* and *vous*, for instance, can be negotiated in some relationships but

tightly policed in institutional ones. They encode social positioning (indexical value), but in many languages they're paradigmatically closed and morphologically integrated. The resulting overlap is expected: a wrong form may be socially sanctionable as address behaviour and structurally repairable as the wrong operator value. This illustrates the channel without collapsing it: an indexical contrast takes on operator value only when it's recruited into the closed-set infrastructure of public update.

8 GRAMMATICALITY JUDGMENTS AS A SOCIAL TECHNOLOGY

If grammaticality talk targets operator value, then grammaticality judgment practices are themselves a social technology: a way of policing the operator repertoire of a communicative situation. Three variables organize this policing.

The **OPERATOR REPERTOIRE** is which contrasts exist and what values they conventionally take: the inventory of operator contrasts available in the community. The **ACTIVATION PROFILE** is which of those contrasts are treated as "in play" in a given register, activity type, or genre: formal written registers may activate operator contrasts (e.g. subjunctive marking) that are dormant in casual speech. The **ACCOUNTABILITY PROFILE** is how strictly departures are treated given participant roles and expectations: L2 learners, children, and visitors to an unfamiliar dialect region may be evaluated against the locally active operator stratum, but participant roles calibrate expectations about competence, affiliation, and repair.

Situation/ascription/identification conditioning gives these profiles their social architecture. The immediate situation selects the activity, medium, footing, and task; speaker ascription supplies durable social baselines such as region, generation, institutional role, or learner status; and identification tracks the norm-centre participants orient to as relevant. These dimensions condition both which operator repertoires are available and how strictly departures are policed. A form can therefore be fully licensed for participants oriented to one norm-centre and non-licensed for participants oriented to another, without either side being confused about its form or meaning (Author, 2026c). Negative concord illustrates the point: *I didn't see nobody* is licensed in some English repertoires and policed in standardized-school repertoires, even when the intended negative meaning is transparent.

These profiles help explain why grammaticality policing exists and what sustains it. Grammatical conventions are community property: they're the stabilized solutions that allow rapid coordination under uncertainty (Clark & Brennan, 1991; Stalnaker, 2002). O'Connor (2019) demonstrates that such coordination problems drive communities to converge on categorical signals that effectively solve local problems but readily entrench arbitrary exclusions. But this coordination story shouldn't be confused with naive instrumentalism. As Pullum (2019) emphasizes, speakers don't have to get grammar right in order to be understood; there's very little pressure to comply for strictly communicative reasons.¹⁰ The relevant commitment is practice-constitutive: participating in a linguistic practice makes speakers answerable to its constraints (Millar, 2004), but this doesn't entail moral obligation, and public accountability is indexed to expected competence.

Repair fits the same pattern. It treats some departures as requiring correction and others as negotiable (Schegloff et al., 1977). Operator mismatches sit at the intersection: they're departures that threaten coordination because they disrupt publicly accountable operator values.

This framing also clarifies why categoricity is common but not inevitable. Some operator contrasts are heavily conventionalized, high-opportunity, and tightly constrained, which makes non-

¹⁰This empirical fact has not noticeably reduced the market for usage advice.

licensing stable and judgments sharp. Others are lower-opportunity or contested across overlapping norm centres, which makes judgments more variable. The key claim is that *wherever categoricity emerges*, it should cluster around operator value.

Speakers frequently label accent or dialect choices as “bad grammar” to enforce social hierarchies. The operator stratum helps distinguish first-order operator mismatches from second-order ideological policing without denying either. Overt correction is rare in both cases, but the target differs. In repair after an operator mismatch, the utterance form is targeted and the update is blocked, distorted, or reclassified: hearers fail to track reference or lack the epistemic anchoring the utterance presupposes. In ideological policing, the speaker is targeted while the update remains intact: the utterance is understood, but the reaction targets the person’s social index (Agha, 2007; Eckert, 2012). The fact that ideologies misappropriate the term “grammar” for the latter doesn’t dissolve the reality of the former.

Slurs and taboo words remain boundary cases for the same reason. They’re publicly accountable in a strong sense, but the typical sanction targets an impermissible action rather than a defective update instruction. The same surface reaction, “you can’t say that”, can therefore block a social move or reject a linguistic instrument.

The analysis addresses what triggers grammaticality talk, not how such talk is socially deployed. The operator stratum provides material for language ideology, but that doesn’t make the stratum ideology-free or judgments politically neutral. Who speaks for “the community”? Who gets to police the repertoire, and with what consequences for whom? Those are questions about the social life of grammaticality talk, not about its target.

The target of grammaticality talk isn’t identical to the output of a judgment task. The target, on this account, is operator value: whether a form is licensed as a public-update instruction in the relevant repertoire. Felt ungrammaticality or unacceptability is detector output: a speaker-level response that can be sharpened or blurred by processing difficulty, familiarity, social penalty, task design, and metalinguistic training. Good detectors track the target often enough to be useful, but they aren’t the target itself. Categorical judgments can therefore be evidence about operator licensing without defining grammaticality as the feeling of ungrammaticality (Author & Coauthor, 2026).

9 PREDICTIONS AND RESEARCH STRATEGIES

The proposal generates tractable empirical predictions across subdisciplines, which is part of its point: it’s meant to be usable by typologists, sociolinguists, psycholinguists, and interaction analysts.

(1) REPAIR ASYMMETRIES. Overt repair is rare overall, but when it does occur, operator mismatches should show a different profile from indexical mismatches: more likely to be formulated as incomprehension or rejection (“what?”, “who did it?”) than as negotiation of stance or appropriateness (Schegloff et al., 1977). Cross-linguistically, the targets of repair should correlate with what’s grammaticalized. Operationally: in annotated conversational corpora, code operator mismatches (tense mismatches, agreement mismatches, switch-reference mismatches) separately from indexical mismatches (accent shifts, register mismatches); the prediction is that operator mismatches show higher rates of open-class repair initiation and explicit rejection, while indexical mismatches show more stance negotiation and accommodation. In naturalistic data, many apparent “operator mismatches” are dialect differences or L2 phenomena, and repair is sensitive to participant roles. The prediction is conditional on controlling for these factors or, more usefully, treating them as part of

the accountability profile theorized in §8.

(2) SATIATION ASYMMETRIES. Repetition should attenuate anomaly responses more readily for non-operator mismatches and for low-entrenchment operator contrasts than for high-opportunity, strongly preempted operator gaps. This follows from the role of opportunity mass in stabilizing non-licensing (Bybee, 2006, 2010), and connects to experimental work on syntactic satiation (Snyder, 2000; Sprouse, 2009). Operationally: in acceptability judgment tasks with repeated exposure, high-opportunity operator mismatches (e.g., subject–verb agreement in obligatory contexts or English left-branch extraction) should resist satiation, while pragmatic oddities and low-frequency operator mismatches should show steeper improvement curves.

(3) PHONOLOGICAL-EXPONENT EFFECTS. In languages where tone or intonation is an operator exponent, operator mismatches should elicit judgment and processing profiles more similar to morphosyntactic anomalies than to accent shifts, consistent with the idea that the relevant factor is operator value, not segmental content (Hyman, 2011; Ladd, 2008). Operationally: compare ERP responses to tonal tense mismatches, segmental agreement mismatches, and sociophonetic accent shifts; the prediction is that the first two pattern together in response profile, potentially including P600-like effects, while the third diverges. It’s a response-profile prediction, not a claim that any single ERP component uniquely diagnoses operator mismatch.

(4) SOCIOLINGUISTIC STRATIFICATION WITHOUT GRAMMATICALIZATION. Accent and register differences should show systematic indexical fields and enregisterment dynamics (Agha, 2007; Eckert, 2008), but the metalinguistic categorization as “grammar” should be predicted to arise mainly when the indexical resource becomes part of a closed operator contrast set (e.g. a grammaticalized honorific agreement paradigm, not merely polite lexical choice).

(5) PROCESSING SIGNATURES AS RESPONSES TO OPERATOR MISMATCH. ERP and related measures should distinguish operator mismatches from conceptual anomalies in ways that are consistent with their different roles in public update, without requiring a modular syntax/semantics division or a one-to-one mapping between mismatch type and component (Friederici, 2011; Kutas & Hillyard, 1980; Osterhout & Holcomb, 1992).

None of these predictions presupposes a particular formal architecture. They presuppose only that communities conventionalize a subset of contrasts as operator contrasts for public update.

10 CONCLUSION

The motivating puzzle was why clausal architecture is treated as “grammar” in much the same way as tense or number marking is, while accent and much lexical choice are treated differently. The answer is that grammaticality talk tracks a difference in value. Expression-shape constraints make utterances recognizable as tokens of the system; payload resources contribute negotiable content or stance; operators are closed-paradigm contrasts that configure public update, allocate participant roles, and constrain uptake.

Clausal architecture belongs to the operator stratum when it packages clause typing, argument linking, dependency management, or cross-clausal coherence. Tense and number contrasts belong there where they’re grammaticalized as operator contrasts. Phonological, gestural, and lexical material can also enter the stratum when they take on the same role. The boundary is functional rather than substance-based.

Operators can attract categorical policing because they behave like protocol headers rather than

payload content. They carry few bits in themselves but cause large entropy reduction downstream, constraining the space of licit interpretations and next moves. A wrong operator value doesn't merely produce a surprising concept combination; it disrupts or reclassifies the mapping from form to public update.

The asterisk is evidence, not a definition. Where it tracks operator mismatch, it marks a missing or mis-set operator value; the judgment reports a speaker's detector response to that licensing failure. The argument claims stable, projectible operator repertoires, not corrective-control mechanisms in the stronger homeostatic sense (Author, 2026b; Weinberger, 2026). The payoff is a cross-linguistic account of why "what counts as grammar" varies with which contrasts a community grammaticalizes, while grammaticality talk still has a principled target.

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